Reverse-check review package — SM_HeavyN_1602_06957

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized .fr alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	model/1602.06957.fr
original model name	SM_HeavyN_1602_06957 (hidden from the agent)
paper	text/1602.06957_source.tex
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the .fr)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LNKin (:=)

```
I/2 N1bar[s1].Ga[v,s1,s2].del[N1[s2],v] - 1/2 mN1 N1bar[s1] N1[s1]
```

LNCCbare (:=)

```
gN/Sqrt[2] * (VeN1 * N1bar.W[m].ProjM[m].e + VmuN1 * N1bar.W[m].ProjM[m].mu + VtaN1 *
 ↪ N1bar.W[m].ProjM[m].ta)
```

LNCC (:=)

```
LNCCbare + HC[LNCCbare]
```

LNNCBare (:=)

```
1/2 * gN/cw * (VeN1 * N1bar.Z[m].ProjM[m].ve + VmuN1 * N1bar.Z[m].ProjM[m].vm + VtaN1 *
 ↪ N1bar.Z[m].ProjM[m].vt)
```

LNNC (:=)

```
LNNCBare + HC[LNNCBare]
```

LNHbare (:=)

```
- gN*mN1/(2*MW) * (VeN1 * N1bar.ProjM.ve H + VmuN1 * N1bar.ProjM.vm H + VtaN1 *
 ↪ N1bar.ProjM.vt H)
```

LNH (:=)

```
LNHbare + HC[LNHbare]
```

LNGbare (:=)

```
I * gN*mN1/(2*MW) * (VeN1 * vebar.ProjP.N1 G0 + VmuN1 * vmbar.ProjP.N1 G0 + VtaN1 *
 ↪ vtbar.ProjP.N1 G0) + I * gN*mN1/(Sqrt[2]*MW) * (VeN1 * ebar.ProjP.N1 GPbar + VmuN1 *
 ↪ mubar.ProjP.N1 GPbar + VtaN1 * tabar.ProjP.N1 GPbar)
```

`LNG (:=)`

```
LNGbare + HC[LNGbare]
```

`LBSM (:=)`

```
LNKin + LNCC + LNNC + LNH + LNG
```

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

The model adds one self-conjugate Majorana fermion N_1. No gauge quantum numbers or indices are declared for N_1, so it is a Standard Model gauge singlet:

$$D_\mu N_1 = \partial_\mu N_1,$$

with no $SU(3)_c$, $SU(2)_L$, or $U(1)_Y$ gauge connection acting on it.

`LNKin`

$$\mathcal{L}_{\texttt{LNKin}} = \frac{i}{2}\,\overline{N_1}\gamma^\mu\partial_\mu N_1 - \frac{1}{2}\,m_{N_1}\,\overline{N_1}N_1.$$

The factors of 1/2 are the standard normalization for a self-conjugate Majorana fermion.

`LNCCbare`

With $P_L \equiv (1-\gamma^5)/2$,

$$\mathcal{L}_{\texttt{LNCCbare}} = \frac{g_N}{\sqrt{2}}\left[V_{eN_1}\,\overline{N_1}\gamma^\mu P_L e\,W_\mu^+ + V_{\mu N_1}\,\overline{N_1}\gamma^\mu P_L \mu\,W_\mu^+ + V_{\tau N_1}\,\overline{N_1}\gamma^\mu P_L \tau\,W_\mu^+\right].$$

`LNCC`

$$\mathcal{L}_{\texttt{LNCC}} = \mathcal{L}_{\texttt{LNCCbare}} + \text{h.c.}$$

Equivalently,

$$\mathcal{L}_{\texttt{LNCC}} = \frac{g_N}{\sqrt{2}}\sum_{\ell=e,\mu,\tau} V_{\ell N_1}\,\overline{N_1}\gamma^\mu P_L \ell\,W_\mu^+ + \frac{g_N}{\sqrt{2}}\sum_{\ell=e,\mu,\tau} V_{\ell N_1}\,\overline{\ell}\gamma^\mu P_L N_1\,W_\mu^-,$$

where the $V_{\ell N_1}$ are real external parameters in the file.

`LNNCBare`

$$\mathcal{L}_{\texttt{LNNCBare}} = \frac{1}{2}\frac{g_N}{c_W}\left[V_{eN_1}\,\overline{N_1}\gamma^\mu P_L \nu_e\,Z_\mu + V_{\mu N_1}\,\overline{N_1}\gamma^\mu P_L \nu_\mu\,Z_\mu + V_{\tau N_1}\,\overline{N_1}\gamma^\mu P_L \nu_\tau\,Z_\mu\right].$$

`LNNC`

$$\mathcal{L}_{\texttt{LNNC}} = \mathcal{L}_{\texttt{LNNCBare}} + \text{h.c.}$$

Equivalently,

$$\mathcal{L}_{\texttt{LNNC}} = \frac{g_N}{2c_W}\sum_{\ell=e,\mu,\tau} V_{\ell N_1}\,\overline{N_1}\gamma^\mu P_L \nu_\ell\,Z_\mu + \frac{g_N}{2c_W}\sum_{\ell=e,\mu,\tau} V_{\ell N_1}\,\overline{\nu_\ell}\gamma^\mu P_L N_1\,Z_\mu.$$

`LNHbare`

$$\mathcal{L}_{\texttt{LNHbare}} = -\frac{g_N m_{N_1}}{2M_W}\left[V_{eN_1}\,\overline{N_1}P_L\nu_e\,H + V_{\mu N_1}\,\overline{N_1}P_L\nu_\mu\,H + V_{\tau N_1}\,\overline{N_1}P_L\nu_\tau\,H\right].$$

`LNH`

$$\mathcal{L}_{\texttt{LNH}} = \mathcal{L}_{\texttt{LNHbare}} + \text{h.c.}$$

Equivalently,

$$\mathcal{L}_{\texttt{LNH}} = -\frac{g_N m_{N_1}}{2M_W}\sum_{\ell=e,\mu,\tau} V_{\ell N_1}\,\overline{N_1}P_L\nu_\ell\,H - \frac{g_N m_{N_1}}{2M_W}\sum_{\ell=e,\mu,\tau} V_{\ell N_1}\,\overline{\nu_\ell}P_R N_1\,H.$$

`LNGbare`

With $P_R \equiv (1+\gamma^5)/2$,

$$\mathcal{L}_{\texttt{LNGbare}} = i\,\frac{g_N m_{N_1}}{2M_W}\left[V_{eN_1}\,\overline{\nu_e}P_R N_1\,G^0 + V_{\mu N_1}\,\overline{\nu_\mu}P_R N_1\,G^0 + V_{\tau N_1}\,\overline{\nu_\tau}P_R N_1\,G^0\right]$$

$$+i\,\frac{g_N m_{N_1}}{\sqrt{2}M_W}\left[V_{eN_1}\,\overline{e}P_R N_1\,G^- + V_{\mu N_1}\,\overline{\mu}P_R N_1\,G^- + V_{\tau N_1}\,\overline{\tau}P_R N_1\,G^-\right].$$

Here `G0` is the neutral Goldstone boson and `GPbar` is the negatively charged Goldstone boson G^-.

`LNG`

$$\mathcal{L}_{\texttt{LNG}} = \mathcal{L}_{\texttt{LNGbare}} + \text{h.c.}$$

That is, it also contains the conjugate G^0 and G^+ interactions implied by hermitian conjugation.

`LBSM`

$$\mathcal{L}_{\texttt{LBSM}} = \mathcal{L}_{\texttt{LNKin}} + \mathcal{L}_{\texttt{LNCC}} + \mathcal{L}_{\texttt{LNNC}} + \mathcal{L}_{\texttt{LNH}} + \mathcal{L}_{\texttt{LNG}}.$$

Field Table

Symbol	Spin	$SU(3)_c$ rep	$SU(2)_L$ rep	$U(1)$ charge / hypercharge	Self-conjugate	Mass
`N1`	1/2	singlet	singlet	0	yes	`mN1 = 300.`

The file declares no color, weak-isospin, or abelian quantum number for `N1`; the interactions therefore treat it as a neutral gauge-singlet Majorana fermion coupled through mixing-like parameters to Standard Model leptons.

Parameters

Symbol	Type	Value	Multiplies	Physical meaning
`VeN1`	external, real	`1.0`	N_1-electron-flavor charged-current, neutral-current, Higgs, and Goldstone terms	Mixing/coupling coefficient between N_1 and electron-flavor leptons

Symbol	Type	Value	Multiplies	Physical meaning
`VmuN1`	external, real	`0.0`	N_1-muon-flavor charged-current, neutral-current, Higgs, and Goldstone terms	Mixing/coupling coefficient between N_1 and muon-flavor leptons
`VtaN1`	external, real	`0.0`	N_1-tau-flavor charged-current, neutral-current, Higgs, and Goldstone terms	Mixing/coupling coefficient between N_1 and tau-flavor leptons

The internal parameter is

$$g_N = \frac{e}{s_W},$$

so the new interactions use the Standard Model weak $SU(2)_L$ coupling normalization.

Physics Summary

The file encodes a single neutral Majorana fermion N_1, a gauge singlet with mass $m_{N_1} = 300$, coupled to Standard Model leptons through flavor-dependent real coefficients V_{eN_1}, $V_{\mu N_1}$, and $V_{\tau N_1}$. It has charged-current interactions with $W^\pm \ell^\mp$, neutral-current interactions with $Z\nu_\ell$, Higgs interactions with $H\nu_\ell$, and the corresponding Feynman-gauge Goldstone interactions. This mediates production and decay channels such as $W^* \to \ell N_1$, $N_1 \to \ell W$, $N_1 \to \nu Z$, and $N_1 \to \nu H$, with Majorana self-conjugacy allowing lepton-number-violating charged-lepton signatures.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Located Paper Definitions

The paper defines the heavy-neutrino model in **Section II, "Heavy Neutrino Model"**. The relevant definitions are:

- **Neutrino mixing / field content:** Section II, first displayed `eqnarray`, approximately **Eq. (7)**:

$$\begin{pmatrix} \nu_{Li} \\ N^c_{Rj} \end{pmatrix} = \begin{pmatrix} U_{3\times 3} & V_{3\times n} \\ X_{n\times 3} & Y_{n\times n} \end{pmatrix} \begin{pmatrix} \nu_m \\ N^c_{m'} \end{pmatrix}.$$

- **Flavor-state expansion:** Section II, next displayed equation, approximately **Eq. (8)**:

$$\nu_\ell = \sum_{m=1}^{3} U_{\ell m}\nu_m + \sum_{m'=1}^{n} V_{\ell m'} N^c_{m'}.$$

- **One-heavy-state simplification:** Section II immediately before the interaction Lagrangian: "For simplicity, we consider only one heavy mass eigenstate, labeled by N."
- **Interaction Lagrangian:** Section II, labeled `\label{eq:Lagrangian}`, approximately **Eq. (9)**.
- **Goldstone implementation:** Section III, "Computational Setup": "We implement the above Lagrangian with Goldstone boson couplings in the Feynman gauge into FeynRules…"

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Heavy state N from one-heavy-mass-eigenstate simplification after Eq. (8), with mixing inherited from RH heavy states N_R^c in Eq. (7)	One self-conjugate Majorana fermion N_1, SM gauge singlet	agree	The paper does not spell out gauge representations in a table, but the RH sterile-neutrino origin implies an SM singlet. Majorana/self-conjugate treatment is consistent with the use of N^c and the heavy-neutrino phenomenology, though the word “Majorana” is not explicit in the extracted text.
Full mixing rotation with U, V, X, Y, Eq. (7)	Only $V_{eN_1}, V_{\mu N_1}, V_{\tau N_1}$ parameters reconstructed	missing-in-reconstruction	The reconstruction captures the active-heavy $V_{\ell N}$ coefficients relevant to the BSM interactions, but not the full neutrino rotation matrix or X, Y blocks.
Flavor state $\nu_\ell = \sum_m U_{\ell m}\nu_m + \sum_{m'} V_{\ell m'} N^c_{m'}$, Eq. (8)	Effective flavor-dependent couplings $V_{\ell N_1}$ to N_1	agree	For the one-heavy-state truncation, the reconstruction captures the active-heavy part. It omits the light-neutrino $U_{\ell m}\nu_m$ piece.
SM light-neutrino charged current: $-\frac{g}{\sqrt{2}} W^+_\mu \sum_{\ell,m} \bar{\nu}_m U^*_{\ell m}\gamma^\mu P_L \ell^-$, Eq. (9), first line	No corresponding term in reconstructed BSM Lagrangian	missing-in-reconstruction	Likely supplied by the base SM model rather than the sanitized BSM extension, but it is present in the paper’s displayed interaction Lagrangian.
Heavy charged current: $-\frac{g}{\sqrt{2}} W^+_\mu \sum_{\ell=e}^{\tau} \overline{N^c} V^*_{\ell N}\gamma^\mu P_L \ell^- +$ H.c., Eq. (9), second line	$+\frac{g_N}{\sqrt{2}} \sum_\ell V_{\ell N_1} \bar{N}_1 \gamma^\mu P_L \ell\, W^+_\mu +$ h.c.	disagree	Field content, chirality, W^+, and $1/\sqrt{2}$ coefficient agree. Differences: paper has an overall minus sign and complex-conjugated $V^*_{\ell N}$; reconstruction uses positive sign and real $V_{\ell N_1}$.
SM light-neutrino neutral current: $-\frac{g}{2c_W} Z_\mu \sum_{\ell,m} \bar{\nu}_m U^*_{\ell m}\gamma^\mu P_L \nu_\ell$, Eq. (9), third line	No corresponding term in reconstructed BSM Lagrangian	missing-in-reconstruction	Again likely part of the base SM/light-neutrino sector rather than the BSM addition, but it is present in the paper’s Eq. (9).
Heavy neutral current: $-\frac{g}{2c_W} Z_\mu \sum_{\ell=e}^{\tau} \overline{N^c} V^*_{\ell N}\gamma^\mu P_L \nu_\ell +$ H.c., Eq. (9), fourth line	$+\frac{g_N}{2c_W} \sum_\ell V_{\ell N_1} \bar{N}_1 \gamma^\mu P_L \nu_\ell Z_\mu +$ h.c.	disagree	The Z, P_L, $1/(2c_W)$, and flavor structure agree. Differences are the same as for the charged current: sign and complex conjugation/reality of V.

paper term (eq. ref)	reconstruction term	verdict	notes
Heavy Higgs interaction: $-\frac{g m_N}{2M_W} h \sum_{\ell=e}^{\tau} \overline{N^c} V_{\ell N}^* P_L \nu_\ell +$ H.c., Eq. (9), fifth line	$-\frac{g_N m_{N_1}}{2M_W} \sum_\ell V_{\ell N_1} \bar{N}_1 P_L \nu_\ell H +$ h.c.	agree	Coefficient, sign, chirality, Higgs coupling, and mass proportionality agree if N_1 is identified with N^c up to Majorana convention and the reconstruction's real $V_{\ell N_1}$ is treated as the paper's $V_{\ell N}^*$ in a real benchmark.
Goldstone couplings in Feynman gauge, Section III	`LNGbare` and `LNG`: neutral G^0 and charged $G^\pm$ mass-proportional couplings with h.c.	agree	The paper states that Goldstone couplings are implemented but does not print their explicit form. The reconstruction's mass-proportional Feynman-gauge Goldstone structure is consistent with the displayed $W/Z/h$ interactions, but exact signs and i conventions cannot be verified directly from the paper text.
Heavy-neutrino free kinetic and mass terms	`LNKin`: $\frac{i}{2}\bar{N}_1 \gamma^\mu \partial_\mu N_1 - \frac{1}{2} m_{N_1} \bar{N}_1 N_1$	extra-in-reconstruction	The paper's displayed Eq. (9) is explicitly the interaction Lagrangian with EW bosons; it does not print the free kinetic/mass term. The term is standard for a Majorana mass eigenstate and not in tension with the paper.
Coupling normalization g, $c_W = \cos\theta_W$, SM inputs Eq. (10)	$g_N = e/s_W$, c_W, M_W	agree	$g_N = e/s_W$ is the standard weak coupling g. The reconstruction's normalization matches the paper's interaction coefficients.
Flavor-mixing parameters $V_{\ell N}$, constrained and cross sections scaling as $\lvert V_{\ell N}\rvert^2$, Section II after Eq. (9)	Real external parameters with defaults $V_{eN_1} = 1$, $V_{\mu N_1} = 0$, $V_{\tau N_1} = 0$	disagree	The paper treats $V_{\ell N}$ as generally model-dependent and uses $\lvert V_{\ell N}\rvert^2$. The reconstruction restricts them to real values and includes benchmark defaults; this is implementation-specific and narrower than the paper's general parameterization.

Disagreements and Checks

- **Heavy charged-current sign and conjugation** — severity: **convention**. A human should check the FeynRules sign conventions and whether the UFO vertices use an overall interaction-sign convention opposite to the paper's displayed Lagrangian.
- **Heavy neutral-current sign and conjugation** — severity: **convention**. A human should check whether $V_{\ell N_1}$ is defined as the real value of $V^*_{\ell N}$, and whether the sign is absorbed by field or Lagrangian conventions.
- **Real-only mixing parameters versus complex $V_{\ell N}$** — severity: **substantive**. A human should check whether the implementation intentionally supports only real active-heavy mixing, since complex phases would matter for CP-sensitive observables.
- **Missing full U, V, X, Y mixing rotation** — severity: **cosmetic** for the heavy-neutrino production model, **substantive** for a full neutrino-sector implementation. A human should check whether the implementation is meant only as a phenomenological one-heavy-state model rather than a complete seesaw mass model.
- **Missing SM light-neutrino charged and neutral currents** — severity: **cosmetic** if inherited from a base SM model, **substantive** if this reconstruction is supposed to describe the complete paper Lagrangian. A human should verify whether those interactions are supplied elsewhere in the model stack.
- **Extra explicit Majorana kinetic/mass term** — severity: **cosmetic**. A human should check only that the implementation's m_{N_1} corresponds to the paper's m_N and that the Majorana normalization is handled consistently.
- **Goldstone coefficient details not printed in the paper** — severity: **convention**. A human should compare directly against the original FeynRules file or generated UFO rules if exact i, charge-conjugation, and Goldstone sign conventions matter.

Overall, the reconstruction captures the central BSM physics of the paper: a single heavy neutral lepton coupled to SM leptons through active-heavy mixing, with the expected W, Z, Higgs, and Feynman-gauge Goldstone interactions and the correct weak-coupling and mass-proportional coefficient structures. The main gaps are that the reconstruction is narrower than the paper's formal neutrino mixing setup, omits the light-neutrino SM pieces shown in the paper's Eq. (9), and treats the active-heavy mixings as real benchmark parameters. The only direct term-level mismatches in the heavy interaction terms are the overall signs and complex conjugation of $V_{\ell N}$ in the charged- and neutral-current interactions, which are likely convention-dependent but should be checked against the actual FeynRules/UFO vertex conventions.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 5 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: